

YOHAKU 2026 · Challenge 7 · Fungi, AI and shared decisions in orbit

the System

Local decisions. A shared safety layer. Human oversight.

Every satellite carries the same small onboard AI. It senses danger locally, announces its plan to its neighbour, decides who moves inside a shared safety layer, asks a human when the stakes are high, and settles the cost of staying safe in a shared ledger. Nothing in the network decides alone.

Live site: <https://maargin.github.io/yohaku-challenge7/> · Code: <https://github.com/maargin/yohaku-challenge7>

The problem

Orbit has no traffic control

207,152

Starlink collision-avoidance manoeuvres reported in six months (Dec 2025 – May 2026)

2019

ESA's Aeolus dodged Starlink 44 alone after a follow-up email went unseen

3,006

real tracked objects on the site's globe today (CelesTrak):
1,500 Starlink, 800 other active, 700 debris, 6 crewed

- Thousands of satellites from different operators share orbit with debris, and nothing decides who moves.
- Today the decision is made by email between operators, hours before the pass, with no shared record.
- As AI takes over these decisions, three questions stay open: who has priority, who pays for the dodge, and who is responsible when it goes wrong.

The solution

Five layers, one decision

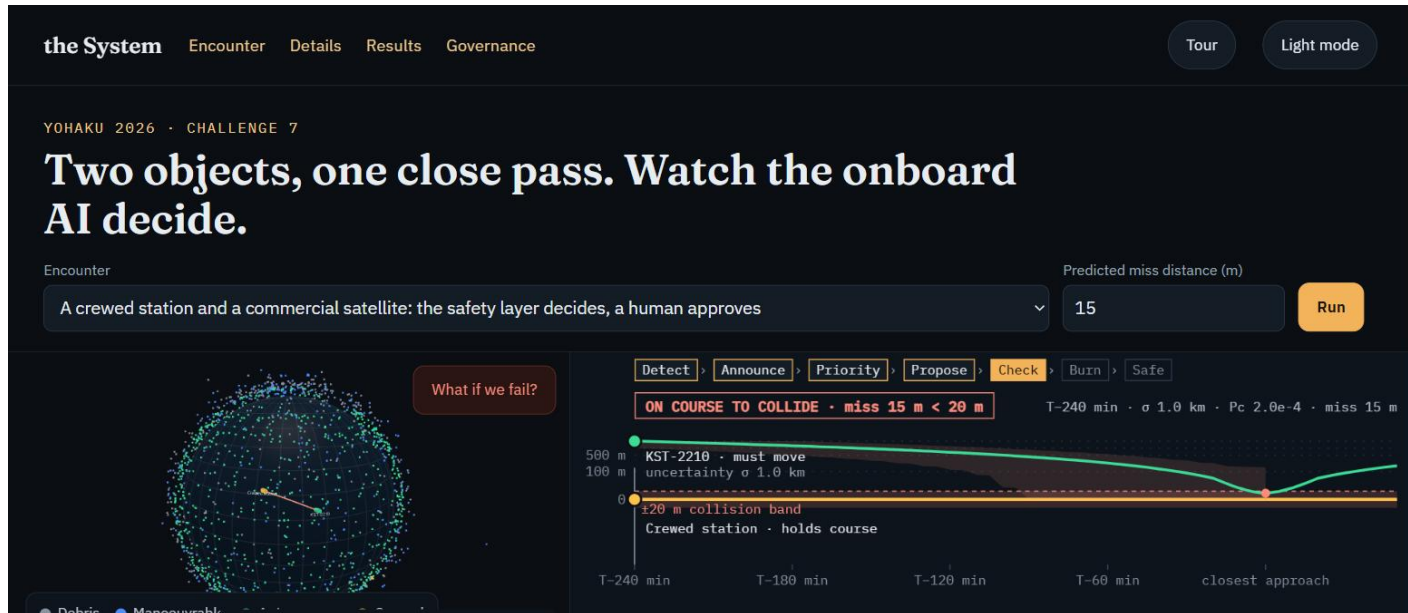
Layer	What it does
Onboard AI	Every satellite runs the same small policy (multi-agent reinforcement learning, 24 inputs → 64 → 64 → 7 actions). It senses its most dangerous neighbour and proposes when and how to burn.
Handshake	PROPOSE / ACK / DO-NOT-MOVE / EXECUTED / ESCALATE replaces email. Both sides see the same data; a party that never answers is treated as unable to move.
Safety layer	A fixed priority check on shared data decides who must move: an object that cannot move never yields; crewed holds; low fuel holds; a free-rider yields; the more capable (autonomous) moves; commercial yields to public good; then a fair tie-break. Eight hard triggers send the case to a human; the AI cannot switch them off.
Human oversight	A crewed vehicle, high risk or a silent counterpart opens a decision: approve, override (only before both sides commit) or stop. Every decision is logged with its actor and time.
Fairness ledger	Double-entry with a commons account: whoever yields earns credit; debris owners, silence and false declarations pay. Balances always sum to zero.

How a decision is made

Seven steps, every time

1 · Sense	Collision probability, tracking uncertainty and predicted miss of the most dangerous neighbour. “Dangerous” means a manoeuvre is needed.
2 · Announce	Both satellites exchange short messages with their registered class and fuel.
3 · Priority check	The safety layer decides who must move, from shared data, in a fixed order.
4 · AI proposal	Each onboard AI proposes one of seven actions with its confidence: hold, small or large opening burn, closing burn, radial burn, request the other to move, escalate.
5 · Safety layer	The proposal is allowed or changed: no burn without priority, closing burns become opening burns, a yielder that waits past T–60 min is made to burn.
6 · Execute	The burn that was actually made, and how the miss distance changes because of it.
7 · Human	When the stakes are high a person approves, overrides or stops; the decision is logged.

Watch the decision, not read about it

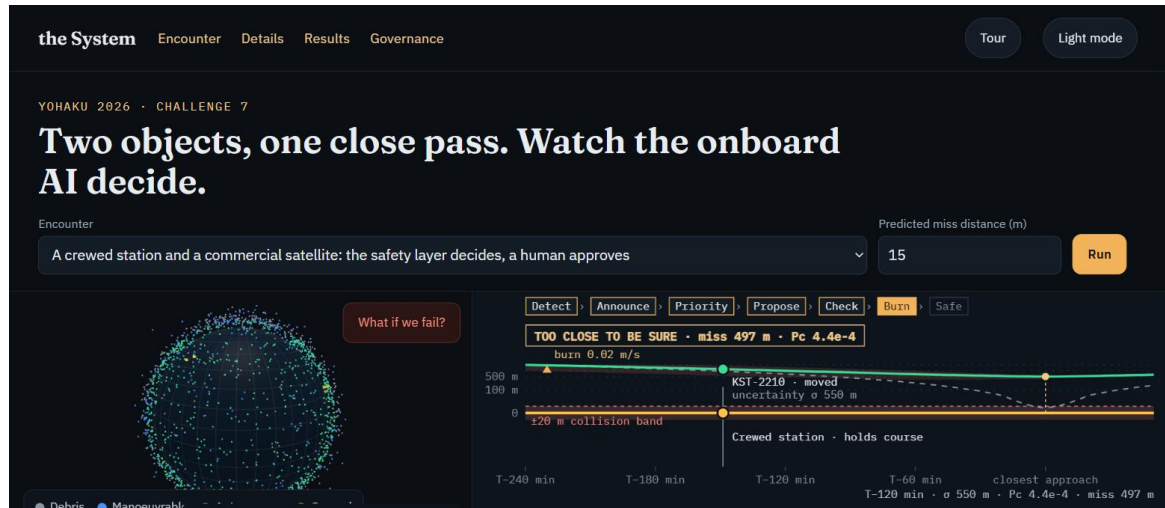


At first warning: the two paths over time, on course to collide, uncertainty still large.

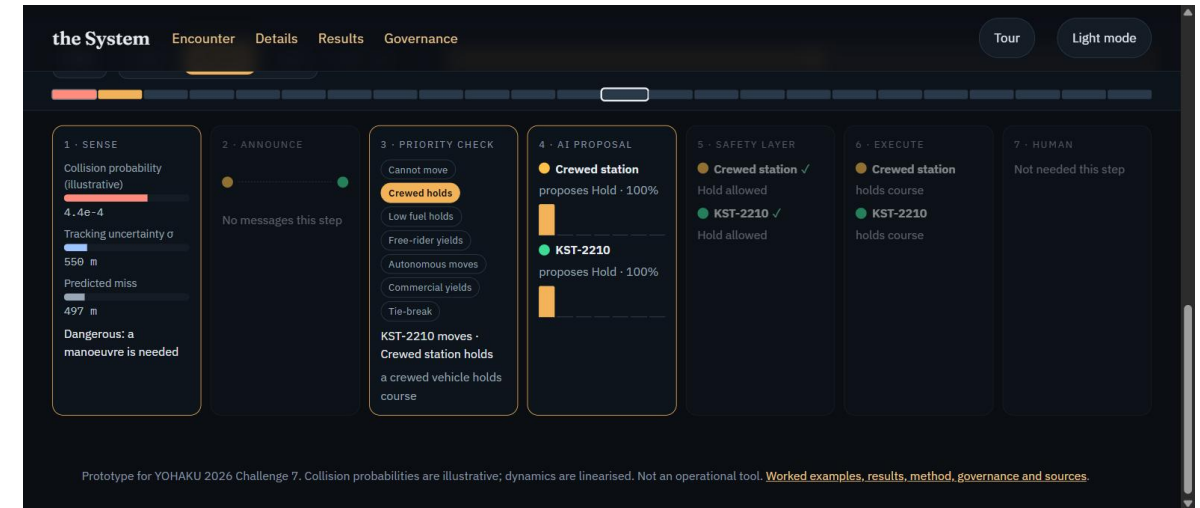
- **Globe:** 3,006 real objects; the two of the encounter are joined by a risk link; the one that burns jumps off the closing line.
- **Approach view:** time runs to closest approach; the object that holds course is a line with a ± 20 m collision band; the other comes in from far away inside an uncertainty funnel; a verdict box says on course to collide / too close to be sure / safe.
- **Timeline:** play, scrub, marks for messages, burns, escalations and decisions.
- **Decision flow:** the seven steps, lit for the current moment.
- **Setup:** presets decided now by the AI, worked examples, or any two real objects from the catalogue.
- **Tour:** a guided walk-through explains every element.

The site · the manoeuvre and the reasoning

Before and after the burn



After the burn: the old path fades, the new one bends outside the band, the verdict changes.



The decision flow at the same moment: which check fired, what each AI proposed, what the safety layer allowed.

Trained by trial and error, kept inside a safety layer

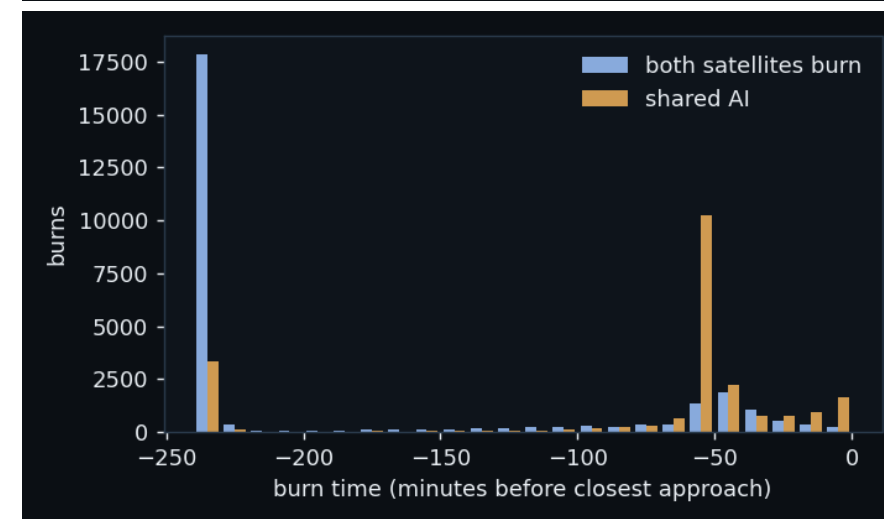
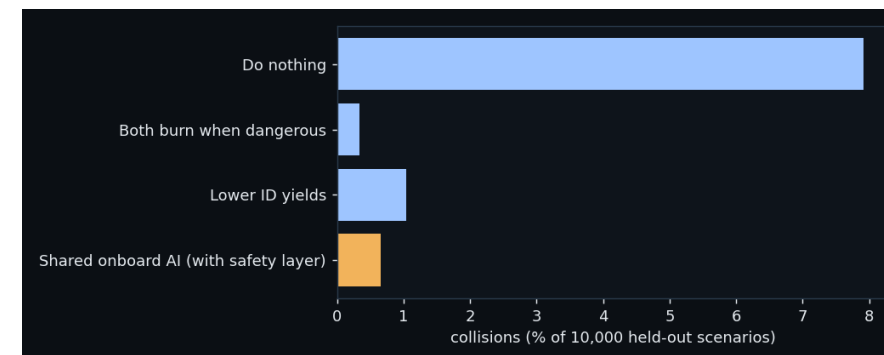
- **Method:** multi-agent reinforcement learning (MAPPO): one shared actor on every satellite, a centralised critic during training only.
- **Environment:** encounters of 2–6 satellites, 24 steps of 10 minutes; tracking uncertainty and miss distances resampled from real ESA Kelvins conjunction events (13,154 events); 30% of scenarios are safe passes so the AI must learn not to burn.
- **Training:** 4,096 encounters in parallel on one GPU; curriculum from debris to mixed traffic with crewed and silent satellites; about 836 million simulated steps per seed.
- **Reward:** collision –100, risk reduction, secondary close pass –20, fuel, fairness, safety-layer conflicts, repeat and unneeded burns. Escalating to a human is never penalised.
- **Explanations:** a local open-weight language model writes the plain-language account from the decision data.
- **Runs anywhere:** the exported network runs in the browser (checked against the trained model, error < 1e-5) and, when the team's simulator server is reachable, every scenario and live encounter is decided by the simulator itself.
- **Live mode:** pick a preset or two real catalogue objects, set the miss distance, and watch the AI decide; open the working of any step: the 24 inputs, the network's probabilities, the physics with its numbers and the safety-layer checks.
- **Tested:** the browser simulator reproduces the GPU environment step by step on recorded encounters; property-based tests cover the safety layer, ledger, handshake and server inputs; 56 front-end and 117 Python tests.

Does the shared AI actually help?

Strategy	Collisions	Mean Δv (m/s)	Manoeuvres / event	Burden Gini
Do nothing	7.91 %	0	0	0
Both burn when dangerous	0.34 %	0.053	2.66	0.20
Lower ID yields	1.04 %	0.030	1.52	0.54
Shared onboard AI (with safety layer)	0.66 %	0.044	2.22	0.43

Metric (95% CI)	Shared onboard AI
Accuracy — scenarios ending safely	99.34 % (99.16 – 99.48)
Recall — would-collide scenarios resolved	91.66 % (89.52 – 93.39)
Precision — burns in would-collide scenarios	7.91 % (7.4 – 8.46)
Collisions (avoidable / unavoidable)	50 / 16

Unavoidable collisions are between two objects that cannot move. Precision is low by nature: under tracking uncertainty every risky pass needs a manoeuvre, and only some would really have collided. Having both satellites burn every time is safer still but uses 20% more fuel and needs no coordination; the trade-off is shown, not hidden.



Who is in charge, who is responsible, what does everyone owe?

When an AI-shaped decision goes wrong, who answers?

- Bad tracking data — the data provider first, the operator second.
- Flawed AI policy — the AI developer, within its declared operating domain.
- Ignored or silent handshake — that operator, presumed responsible.
- False class declaration — the declaring operator; the licensing state supervises.
- Human override that causes harm — the operator whose person decided; every decision is logged with its actor.
- Evidence of fault — whether the safety layer was respected, and what the ledger and decision log show.

The network learns. People decide what it learns. Every case logged → update proposed → an independent Orbital Incident Board approves → the new version is recorded. Neither the safety layer nor the model changes by itself.

NARETU obligations charter

- **Operators owe** shared data, a timely handshake, their fair share of dodges, and disposal at end of life.
- **AI developers owe** explainable decisions, a bounded operating domain, and triggers they cannot switch off.
- **Data providers owe** neutral, timely warnings with honest uncertainty.
- **States owe** supervision of the operators they license, and verified registration.
- **Everyone owes those not yet launched** a usable orbit: no net addition to long-lived debris, and protection for the small, the stuck and the crewed first.

NARETU framing from the challenge brief, credited to Chief Titus Letaapo and Samburu obligations-based governance.

What we borrowed, what we did not, what comes next

From the challenge's mycelium lens we borrowed the architecture: local sensing, a connected network, and rewards for partners that give back. We did not borrow its morality: fungi cannot be held responsible and repair themselves; orbit cannot. So the system adds what nature lacks: priority, accountability and a human veto.

Limits. Dynamics are linearised and collision probabilities are illustrative; this is not an operational tool. Scenarios are synthetic encounters built from real uncertainty and miss statistics. The 2019 replay is scripted from ESA's published timeline.

Next steps.

- Operators and industry groups: publish the handshake as an open message specification on top of existing ephemeris sharing.
- Coordination services: pilot the shared ledger as a read-only record of who yielded.
- Regulators and standards bodies: use the safety layer's priority order as a starting point for harmonised right-of-way standards.
- Universities: run the simulator as a testbed; ask operators, students and the public which priorities they accept.

Data and software. CelesTrak (T.S. Kelso) orbital elements and SOCRATES screening; ESA Kelvins Collision Avoidance Challenge dataset (CC-BY-4.0); ESA's 2019 account; SpaceX FCC reports. globe.gl, three.js, satellite.js, Chart.js; Fraunces and IBM Plex fonts.